

Homework 3

requirements:

- Phase margin must be at least 60°
- The settling time of a step response within a $\pm 2\%$ band must be less than 3 seconds, while
- the steady state error of the system in the case of a unit ramp input should not exceed 0.1.
- the bandwidth of the system not exceed 15 rad/s

```
clearvars, clc, clear all
```

$$G(s) = \frac{50(s+15)}{s^4 + 36s^3 + 484s^2 + 3280s}$$

```
num = 50 * [1 15];  
den = [1 36 484 3280 0];  
G = tf(num, den);  
G
```

G =

$$\frac{50 s + 750}{s^4 + 36 s^3 + 484 s^2 + 3280 s}$$

Continuous-time transfer function.
Model Properties

```
Gzpk = zpk(G)
```

Gzpk =

$$\frac{50 (s+15)}{s (s+20) (s^2 + 16s + 164)}$$

Continuous-time zero/pole/gain model.
Model Properties

the steady state error is calculated in terms of the ramp error constant K_v and thus a required value of is established:

$$e_{ss} = \frac{A}{K_v} \leq 0.1$$

$$K_v \geq 10$$

It is important to realize that this value represents the required K_v .

The actual K_v will depend on the gain chosen by the designer. If for instance the designer chooses to start his design by selecting $D_c(s) = 1$ and he/she then calculates the actual K_v . Here $K_v = \frac{50 \times 15}{20 \times 164} = \frac{75}{328}$, is small by a factor of 43.73. So we can put a gain $K_1 = 44$ in series with the plant.

$K_1 = 44$

$K_1 =$
44

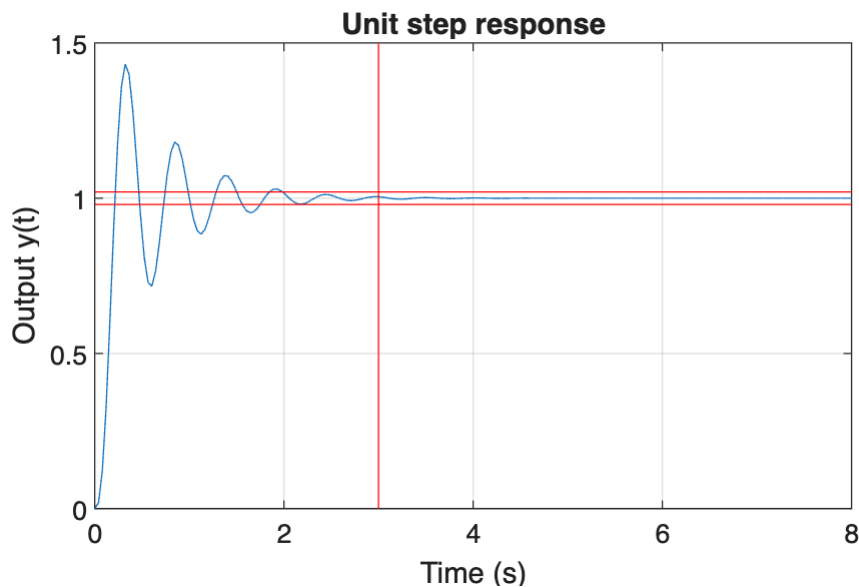
$T = \text{feedback}(K_1 * G, 1)$

$T =$

$$\frac{2200 s + 33000}{s^4 + 36 s^3 + 484 s^2 + 5480 s + 33000}$$

Continuous-time transfer function.
Model Properties

```
tt = linspace(0.0,8.0,200);
y = step(T,tt);
figure(1)
plot(tt,y)
hold on
plot([tt(1), tt(end)], [1.02, 1.02], 'r')
plot([tt(1), tt(end)], [0.98, 0.98], 'r')
plot([3,3], [0,1.5], 'r')
grid on
xlabel('Time (s)')
ylabel('Output y(t)')
title('Unit step response')
```



Now we plot the Bode diagram of the uncompensated system.

```

figure(2)
omega = logspace(0,2,400);
[mag1,phase1] = bode(K1*G,omega);
mag1 = squeeze(mag1);
phase1 = squeeze(phase1);
subplot(2,1,1),semilogx(omega,20*log10(mag1))
xlabel('Angular frequency \omega (rad/s)'),ylabel('20log|G(\omega)|')
hold on
grid on
title('Plant (open loop) transfer function bode diagram')
subplot(2,1,2),semilogx(omega,phase1)
xlabel('Angular frequency \omega (rad/s)'),ylabel('\phi')
hold on
grid on
[Gm,Pm,omegcp,omegcp] = margin(K1*G);
dBGm = 20*log10(Gm);
disp(['System phase margin (degrees): ',num2str(Pm)])

```

System phase margin (degrees): 27.24

```
disp(['System cross over angular frequency (rad/s): ',num2str(omegcp)])
```

System cross over angular frequency (rad/s): 10.2164

```
disp(['System gain margin (dB): ',num2str(dBGm)])
```

System gain margin (dB): 4.6566

```
disp(['System Phase Crossover Frequency (rad/s):',num2str(omegcp)])
```

System Phase Crossover Frequency (rad/s):13.9847

We therefore need to turn to step 3: Determine the frequency where the phase margin requirement would be satisfied if the magnitude curve crossed the 0 dB line at this frequency, ω_c' , (Allow for 5° phase lag from the phase-lag compensator when determining the new crossover frequency.)

Since the phase margin of the gain we need is 60° , It is suggested that we add a measure of safety of 5 degree, so I decided to start off my design iteration with the value of ω_c' at the which PM is $-180^\circ + 60^\circ + 5^\circ = -115^\circ$

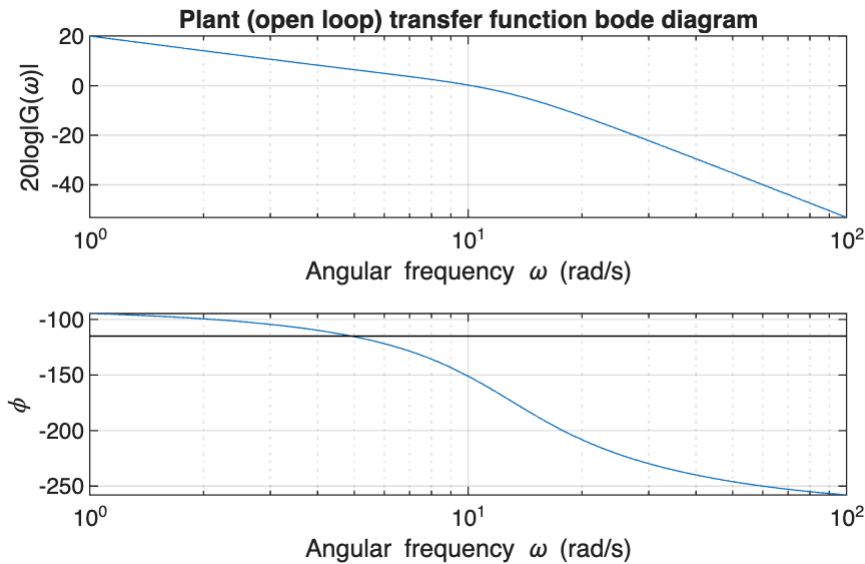
From Franklin Eq.6-40

$$\alpha = \frac{1 - \sin \phi_{\max}}{1 + \sin \phi_{\max}}$$

```

target_phase = -115;
figure(2)
subplot(2,1,2),semilogx([omega(1),omega(end)],
[target_phase,target_phase],'k')

```



```
wc_prime = interp1(phase1, omega, target_phase);
T_I = 10/wc_prime
```

```
T_I =
2.0357
```

```
disp(['omega_c_prime at target PM is: ', num2str(wc_prime)])
```

```
omega_c_prime at target PM is: 4.9124
```

```
mag_wc_prime = interp1(omega, mag1, wc_prime);
attenuation_dB = -20 * log10(mag_wc_prime);
disp(['necessary attenuation at \omega_c_prime to ensure curve crosses 0
dB is : ', num2str(attenuation_dB)])
```

```
necessary attenuation at \omega_c_prime to ensure curve crosses 0 dB is : -6.6058
```

```
alpha = mag_wc_prime
```

```
alpha =
2.1394
```

From the figure, the ω_c' is read by 4.912 rad/s,

$$z = \frac{1}{T_I} = \frac{\omega_c'}{10} = 0.491, T_I = 2.0357$$

When $\omega_c' = 4.912$, $20\log(G(\omega)) = -6.6058$

$$\text{attenuation} = -20\log_{10}(\alpha) \Rightarrow \alpha = 2.1394$$

and then the compensator is calculated as: $D_C = \frac{T_I s + 1}{\alpha T_I s + 1}, \alpha > 1$

```
Dc = tf([T_I, 1], [alpha*T_I, 1])
```

Dc =

$$\frac{2.036 s + 1}{4.355 s + 1}$$

Continuous-time transfer function.
Model Properties

```
disp("Lag compensator zero = -"+num2str(1/T_I)+"rad/s")
```

Lag compensator zero = -0.49124rad/s

```
disp("Lag compensator pole = -"+num2str(1/(alpha*T_I))+"rad/s")
```

Lag compensator pole = -0.22962rad/s

```
disp('Compensated loop transfer function:')
```

Compensated loop transfer function:

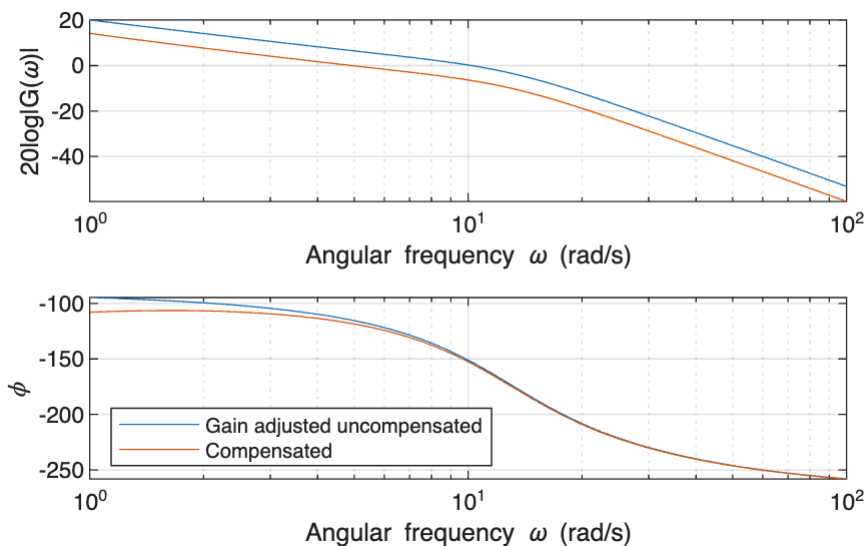
```
Gt = Dc*K1*G
```

Gt =

$$\frac{4478 s^2 + 6.938e04 s + 33000}{4.355 s^5 + 157.8 s^4 + 2144 s^3 + 1.477e04 s^2 + 3280 s}$$

Continuous-time transfer function.
Model Properties

```
figure(3)
title('Loop transfer functions Bode diagrams')
[mag2,phase2] = bode(Gt,omega);
mag2 = squeeze(mag2);
phase2 = squeeze(phase2);
subplot(2,1,1)
semilogx(omega,20*log10(mag1))
hold on
semilogx(omega,20*log10(mag2))
hold off
grid on
xlabel('Angular frequency \omega (rad/s)'),ylabel('20log|G(\omega)|')
subplot(2,1,2)
semilogx(omega,phase1)
hold on
semilogx(omega,phase2)
hold off
grid on
xlabel('Angular frequency \omega (rad/s)'),ylabel('\phi')
legend('Gain adjusted uncompensated','Compensated','Location','southwest')
```



```
[Gm,Pm,omegcp,omegcg] = margin(Gt);
dBGm = 20*log10(Gm);
disp(['Compensated system phase margin (degrees): ',num2str(Pm)])
```

Compensated system phase margin (degrees): 61.8577

```
disp(['Compensated system cross over angular frequency (rad/s): ',...
num2str(omegcg)])
```

Compensated system cross over angular frequency (rad/s): 4.9325

```
disp(['Compensated system gain margin (dB): ',num2str(dBGm)])
```

Compensated system gain margin (dB): 11.0383

```
disp(['Compensated system gain margin cross over angular frequency (rad/s):
'...
,num2str(omegcp)])
```

Compensated system gain margin cross over angular frequency (rad/s): 13.8149

```
disp("Compensated system transfer function"+...
"(i.e., the closed loop transfer function):")
```

Compensated system transfer function(i.e., the closed loop transfer function):

```
T2 = feedback(Gt,1)
```

T2 =

$$\frac{4478 s^2 + 6.938e04 s + 33000}{4.355 s^5 + 157.8 s^4 + 2144 s^3 + 1.925e04 s^2 + 7.266e04 s + 33000}$$

Continuous-time transfer function.
Model Properties

```
y2 = step(T2,tt);  
figure(4)  
plot(tt,y,tt,y2)  
legend('Gain adjusted uncompensated','Compensated','location','southeast')  
hold on  
plot([tt(1), tt(end)], [1.02, 1.02], 'r')  
plot([tt(1), tt(end)], [0.98, 0.98], 'r')  
plot([3, 3], [0.0, 1.5], 'r')  
grid on  
xlabel('Time (s)')  
ylabel('Output y(t)')  
title('Unit step response')  
hold off
```

